

MatDataHub Admin Guide: Adding Materials

1. Introduction

This guide explains how to add new materials to the MatDataHub database. As an administrator, you can add materials individually using the form, or in bulk by uploading a CSV or Excel (.xlsx) file.

2. Single Material Add

Navigate to the "Manage Materials" tab in the Admin Dashboard. Fill in the necessary fields such as Name, Category, Description, and any relevant mechanical or thermal properties. Blank fields will be ignored and stored as empty in the database, allowing for incomplete data entry. Click "Create Material" when done.

3. Bulk Uploading (CSV / Excel)

To upload multiple materials at once, prepare a spreadsheet (.csv or .xlsx). The column headers in the first row **MUST** match the internal database property names exactly.

4. Accepted Column Headers

Required/Common fields:

- name (e.g., "Inconel 718")
- category (e.g., "Metals", "Polymers")
- subcategory (e.g., "Superalloys", "Thermoplastics")
- description
- standard
- grade

Properties (Numeric values only. Leave blank if unknown):

- yield_strength_min, yield_strength_max
- tensile_strength_min, tensile_strength_max
- elastic_modulus
- hardness_value
- density
- max_service_temp
- thermal_conductivity
- specific_heat
- melting_point
- embodied_carbon
- water_usage
- cost_per_kg_min

Note: If a property is not mentioned or left blank in the spreadsheet, it will remain blank in the database.

MatDataHub Admin Guide: Adding Materials

5. Automating Material Prices

To keep prices updated and build user trust, material prices can be automated via scheduled scraping or API integration (e.g., LME or COMEX). This automation periodically fetches latest prices and inserts them into the PriceHistory table, ensuring historic traceability and updating the visible cost_per_kg_min field. This provides transparency to the users that the prices are up to date.